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Source: *Ethnomusicology*, Vol. 36, No. 2 (Spring - Summer, 1992), pp. 171-195

Published by: [University of Illinois Press](#) on behalf of [Society for Ethnomusicology](#)

Stable URL: <http://www.jstor.org/stable/851913>

Accessed: 20/06/2014 22:35

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Modulation in Arab Music: Documenting Oral Concepts, Performance Rules and Strategies

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Although modulation (the practice of moving from one *maqam* to another within a given piece of music) is widely recognized as one of the most important aspects of Arab modal practice, the subject has been all but ignored in written Arabic music theory.¹ With few exceptions, theorists over the centuries have concentrated on describing the modes as individual and distinct entities rather than discussing movement among the modes. In keeping with this lack of attention in theoretical treatises, there is no universally recognized Arabic term for modulation. When discussing modulation, writers and musicians rely on any of three common verbs and their verbal nouns: *yantaqil/intiqāl* ("to shift, change locality, to move"/"change of locality, relocation, transfer"); *yuhawwil/tahwīl* ("to transfer, change switch"/"transformation, change, translocation, transfer"); *yughayyir/taghyīr* (pronounced *taghīr*) ("to change"/"change").²

Armed with methodologies new and old, the modern scholar can play an important role by documenting—in many respects, for the first time—this important aspect of Arab musical practice. Indeed, the etically based scholar is at an advantage in this endeavor since he or she might be less inclined to accept the indigenous boundaries of written theoretical scholarship; that is, the preponderant relegation of the issue of modulation to the realm of music practice and oral theory rather than written theory.

The music and Arabic-language theoretical works under consideration are from the eastern Mediterranean region ranging from Egypt to Syria and Lebanon. In this article, I will address four points based largely on fieldwork conducted in Cairo, Egypt, but also taking into consideration the few modern-day writers who have addressed these issues. After affirming the central importance of modulation in Arab music, I will discuss classificatory concepts which exist most commonly in oral rather than written realms, rules

which govern current modulatory practice, and performers' strategies for executing specific modulations.

The Central Role of Modulation in Arab Music

Although the subject has not been addressed in most theoretical works, there is sufficient evidence to show that modulation has played a major role in Arab music performance from the medieval period to the present day. For example, basing his discussion on thirteenth-century treatises including the *Durrat al-Taj* (c. 1300), Owen Wright describes the practice at that time: "Although a composition would generally be based on just one mode, extraneous units could also be judiciously introduced, especially in improvisatory passages displaying to the full the performer's technical prowess" (1974:498; see also a similar statement in Wright 1980:519). Amnon Shiloah concludes that a group of subsidiary modes in the medieval and pre-modern practice (the *shu'ab* or *murakkabāt*) were probably "not complete or independent modes, but rather serve[d] towards the elaboration of the principal modes" (1981:37). G. A. Villoteau quotes from a medieval or late-medieval treatise which presents the same understanding: "As for the other [modes], many are never used. . . . Nevertheless, one employs some of their notes in the composition of the [main modes], and they produce there an agreeable effect" (1826:68). Shiloah also describes a treatise from 1329 as containing "advice concerning the passage from one mode to another" (1979:249). And finally, Wright mentions a compositional genre, apparently from the fourteenth to sixteenth centuries, called *kull al-nagham* (literally, "all of the modes") in which "a progression through different modes resulted in the inclusion of all the seventeen notes of the octave gamut" (1974:499).

Comments in nineteenth- and twentieth-century sources confirm the continued importance of modulation throughout the modern period.³ Villoteau writes, "Following that which we have observed in the [performance] practice [in Egypt from 1798–1801], each mode can receive . . . some of the notes characteristic of other modes" (1826:126). He also records the existence of "rules for passing from one mode to another" (ibid.:126–27). Mikhā'il Mashāqah, a Lebanese scholar familiar with Egyptian and Syrian musical practice, ignores modulation except for the following passage: "A melody . . . [can] be in a variety of modes to the extent that . . . nothing forbids the use of all the musical modes. Sometimes [a musician] begins in a mode, then moves from it to another mode [showing his] mastery of the art, then returns to [the first mode] at the final repose" ([c.1840]1899:1081; 1913:115).

More recent comments attest to the prevalence and essential nature of modulation in the twentieth century. Jules Rouanet, for example, writes, "It

is quite rare, except in [short pieces], that all of a melody remains in only one mode" (1922:2767). A number of writers, in fact, state that changes in maqam are absolutely necessary. "If we compose a piece in [any one mode] without mixing it with others the listener would be bored" (al-Anṭūnī 1925:3; see also al-Shawwā 1946:96). Describing the *taqāsīm* genre, Samha El-Kholy, an Egyptian scholar, remarks that "to avoid monotony, a change of mode becomes imperative" (1978:19).

Some even argue that modulations represent the most important aesthetic moments in Arab music's modal practices. El-Kholy, for example, considers the modulatory sections of a *taqāsīm* to be "the climax of the [*taqāsīm*]," "the high point of the [*taqāsīm*] structure" (1978:20–21). Edith Gerson-Kiwi writes that modulation "seems to be the vital point in the technique of *maqām* composition without which no artistic level could be reached." She adds, "[I]f done well, nothing can equal the aesthetic pleasure of the listener in the detection of the smooth soldering junctions between [different maqamat]" (1970:71, 72).

Indeed, modulation plays an important role by helping to define the structure of many compositional genres. The instrumental genres, *samā'ī* and *bashraf*, for example, have four sections (*khānāt*, sing. *khānah*) which are each separated by a refrain (*taslīm*) (thus, A X B X C X D X, X being the refrain). The modal structure for these genres commonly has the first section and the refrain set in the original maqam; the second, third, and often the fourth sections are then set in different maqamat. See, for example, Example 1, "*Samā'ī* Bayyātī al-Thaqīl" (for similar, but not identical transcriptions of this piece, see 'Arafah and 'Ali 1984:29; Fataḥ Allāḥ and Kāmil 1982:105) in which the first section and *taslīm* are set in maqam Bayyātī, while the second, third, and fourth sections are in the maqamat Rāst Nawá, Ḥijāz, and Šabá, respectively. People composing new *samā'īyyāt* today still follow this basic modulatory formula.

Figure 1: Scales of the modes used in "Samā'ī Bayyātī al-Thaqīl":

Bayyātī	D	E♭	F		G	A	B♭	c	d	
Rāst Nawá					G	A	B♭	c	d	e f [≠] g
Ḥijāz	D	E♭		F♯	G	A	B♭	c	d	
Šabá	D	E♭	F	G♭		A	B♭	c	d♭	

Modulations often play structural roles in vocal compositions also. In the Umm Kulthūm song "Rubā'iyāt al-Khayyām" composed by Riyad al-Sinbatī (Sono Cairo cassette 05), for example, the dominant maqam is Rāst (and Sūznāk) but there are also large sections in Sūznāk, Nahāwand, Nawá Athar, Bayyātī Nawá, Ḥijāz Nawá, and Sīkāh/Huzām.

Example 1: Samā'ī Bayyātī al-Thaqīl, composer unknown (after a transcription by A. Jihad Racy)

1st khānah ①

③

taslīm ⑤

⑦

2nd khānah ⑧

⑩

3rd khānah ⑫

⑭

4th khānah ⑮

⑰

⑱

⑳

Figure 2. Scales of the modes used in “Rubā’iyāt al-Khayyām”:

Rāst	C	D	E $\flat$	F	G	A	B $\flat$	c				
Sūznāk	C	D	E $\flat$	F	G	A $\flat$	B	c				
Nahāwand	C	D	E $\flat$	F	G	A $\flat$	B	c				
Nawā Athar	C	D	E $\flat$	F $\sharp$	G	A $\flat$	B	c				
Bayyātī Nawā					G	A $\flat$	B $\flat$	c	d	e $\flat$	f	g
Ḥijāz Nawā					G	A $\flat$	B	c	d	e $\flat$	f	g
Sikāh			E $\flat$	F	G	A	B $\flat$	c	d	e $\flat$		
Huzām			E $\flat$	F	G	A $\flat$	B	c	d	e $\flat$		

The importance of modulation is also recognized in statements which present modulation as one of the primary ways for a musician or composer to exhibit his intellectual and technical mastery of his art. Wright (quoted above) notes this function for modulation in the thirteenth century. Discussing the present-day practice, Shiloah writes, “A skilled musician distinguishes himself by excelling in the modulatory progressions that allow him to move away from and back to the main maqam” (1981:40). The Egyptian music historian, Maḥmūd Kāmil, adds that *taqāsīm* is considered a test of, among other things, a performer’s adeptness at modulating from one maqam to another (1975:47; see also Gerson-Kiwi 1970:72).

During fieldwork in Cairo (including music lessons on the *‘ūd* and *nāy*), I found that musicians consistently acknowledged the central role of modulation. A number of my informants stressed this point by saying that the maqamat do not stand by themselves, isolated from one another. Rather, they exist in a state in which they “overlap each other” (that is, they are “*mutadākḥilab*”) (personal communications, Qadri Surūr, 8 June 1987, and Mansi Amin, 13 June 1987). This was especially apparent in music lessons. Lessons, each roughly an hour in length, were often devoted to a single maqam. However, after concentrating for the first ten minutes on phrases within the chosen maqam, my teachers commonly spent the remaining fifty minutes explaining and demonstrating modulations that could be performed from this mode. Clearly, my teachers conceptualized more about modulation than about movement within a given maqam.⁴

From these lessons, I learned that each maqam is part of a fabric that includes all the maqamat (or at least a large number of neighboring maqamat). To know any one maqam fully, a student must know all the places to which one can modulate. This stands in marked contrast to Indian music, where the *rāgas* are understood to exist independently. In North Indian music, it is commonly felt that a student can spend three to five years learning a given *rāga* and, in a sense, master it without having studied a second *rāga*. This is not the case in Arab music, where to master one maqam is to master virtually all the maqamat.

Concepts which Serve to Classify the Modulations that Occur in Practice

A native and largely oral typology of the modulations that occur in practice is found in a number of binary labels mentioned by present-day musicians and theorists. The most objective of these compares the tonic of the original *maqam* with that of the new mode. Modulations are then labeled either as “tonic modulations,” when the new mode shares the same tonic as the original mode, or as modulations to a degree other than the tonic.⁵ In interviews, contemporary theorists recognize two types of tonic modulations: those to modes which share the same lower tetrachord (or pentachord)—the change occurs only in the upper tetrachord (see fig. 3)—and those to modes which have a new lower tetrachord (see fig. 4). In the modern-day theory, modes which share a common lower tetrachord, but differ in the makeup of their upper tetrachord, are said to be from the same *faṣīlah* (“family, genus, species”; plural, *faṣā’il*).⁶

Figure 3: Examples of modes of the same *faṣīlah* among which tonic modulations are possible:

	lower tetrachord				upper tetrachord			
a. Rāst:	C	D	E $\flat$	F	G	A	B $\flat$	c
Sūzdulār:	C	D	E $\flat$	F	G	A	B $\flat$	c
Sūznāk:	C	D	E $\flat$	F	G	A $\flat$	B	c
Nīrz:	C	D	E $\flat$	F	G	A $\flat$	B $\flat$	c
Māhūr:	C	D	E $\flat$	F	G	A	B	c
Dalanshīn:	C	D	E $\flat$	F	G	A	B $\flat$	c d $\flat$
	lower tetrachord				upper tetrachord			
	lower tetrachord				upper tetrachord			
b. Bayyātī:	D	E $\flat$	F	G	A	B $\flat$	c	d
Shūrī:	D	E $\flat$	F	G	A $\flat$	B	c	d
Bayyātayn:	D	E $\flat$	F	G	A $\flat$	B $\flat$	c	d
	lower pentachord				upper tetrachord			
c. Nawā Athar:	C	D	E $\flat$	F $\sharp$	G	A $\flat$	B	c
Nakrīz:	C	D	E $\flat$	F $\sharp$	G	A	B $\flat$	c
Basandīdah:	C	D	E $\flat$	F $\sharp$	G	A	B $\flat$	c

Figure 4: Examples of modes of different faṣā'il among which tonic modulations are possible:

	lower tetrachord				upper tetrachord			
a. Rāst:	C	D	E $\flat$	F	G	A	B $\flat$	c
Nahāwand:	C	D	E $\flat$	F	G	A $\flat$	B	c
Nawā Athar:	C	D	E $\flat$	F $\sharp$	G	A $\flat$	B	c
	lower pentachord				upper tetrachord			
b. Bayyātī:	D	E $\flat$	F	G	A	B $\flat$	c	d
Hijāz:	D	E $\flat$	F $\sharp$	G	A	B $\flat$	c	d
Kurd:	D	E $\flat$	F	G	A	B $\flat$	c	d
Sabā:	D	E $\flat$	F	G $\flat$	A	B $\flat$	c	d $\flat$
	lower tetrachord				upper tetrachord			

In practice, modulations to a degree other than the tonic are most commonly to the note that starts the original mode's upper tetrachord. This note, called the *ghammāz* in present-day Arab music theory, is G for most modes (for example, for Rāst, Nahāwand, Nawā Athar, Nakrīz, Bayyātī, Hijāz, Kurd, Sikāh, and Huzām). For maqām Šabā and 'Ajām (or 'Ajām 'Ushayrān) this note is F, while for a few C-based modes this note is either F or G (for example, for Hijāz Kār and Hijāz Kār Kurd).

Modulations to other notes are also possible. The choice of notes is based on the degree of "compatibility" understood to exist between the various notes and the tonic pitch. The range of possibilities has been addressed in the existing Arab music theory in two ways. Sāmī al-Shawwā, a Syrian born, Cairo-based violinist, and one of the most famous Arab violinists of the first half of the twentieth century, developed a theory of "relative notes" (*aqārib*) in which he indicated that one can modulate to maqāmat based on the original mode's fourth, fifth, and, in some cases, third and sixth degrees (1946:96–99). A number of present-day theorists also deal with this issue by expanding the concept of *ghammāz* to allow for *ghammāzāt*, or multiple *ghammāzs* (see Marcus 1989:552–67). As some examples of *ghammāzāt*, Qadrī Surūr, a Cairo-based musician, theorist, and music educator, mentions the notes G, A, and E half-flat for maqām Rāst. By doing so, Surūr accounts for modulations from Rāst to Bayyātī on G (Bayyātī Nawā). Šabā on A, and Sikāh/Huzām on E half-flat.⁷ For maqām 'Ajām 'Ushayrān, he mentions D and F as *ghammāzāt*, thereby acknowledging modulations to Šabā on D and to Hijāz on F (see fig. 5). As for notes that are considered "incompatible," or

Figure 5: Modulations to modes based on degrees other than the tonic (i.e., modulating to the *ghammāzāt*):**Modulations from Rāst:**

Rāst	C	D	E $\flat$	F	G	A	B $\flat$	c									
Huzām on E $\flat$			E $\flat$	F	G	A $\flat$	B	c	d	e $\flat$							
Bayyātī Nawā					G	A $\flat$	B $\flat$	c	d	e $\flat$	f	g					
Ṣabā on A						A	B $\flat$	c	d $\flat$	e	f	g	a $\flat$				

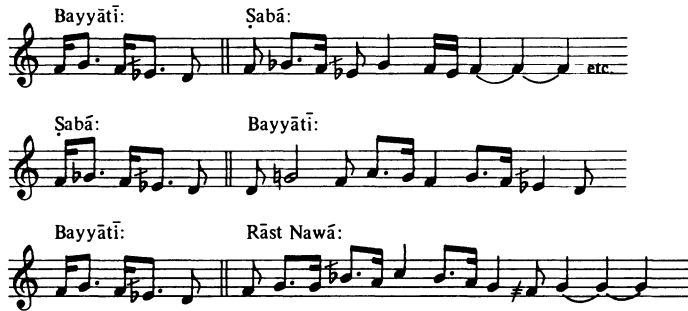
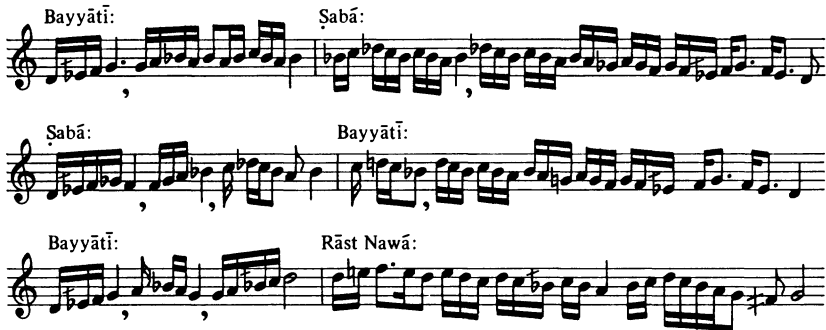
Modulations from ‘Ajām ‘Ushayrān:

‘Ajām ‘Ush. B $\flat$	C	D	E $\flat$	F	G	A	B $\flat$										
Ṣabā on D			D	E $\flat$	F	G $\flat$	A	B $\flat$	c	d $\flat$							
Hijāz on F					F	G $\flat$	A	B $\flat$	c	d $\flat$	e $\flat$	f					

not capable of becoming the tonic note of a new mode, most of my Cairo informants mention the second and seventh degrees of a modal scale. (These notes are also omitted from al-Shawwā’s *aqārib*.) For example, one does not modulate from the C-based maqām Rāst to a mode based on D, B half-flat, or B $\flat$.⁸ The issue of modulating to degrees other than the tonic is discussed again later in this article.

Present-day musicians and music teachers have mentioned at least three other ways to classify the modulations. These differ from the tonic/non-tonic classification in that they all involve a degree of subjectivity and are thus open to personal interpretation. The first, used by Jihad Racy in his classes at UCLA and by two of my ‘ūd teachers in Cairo, recognizes sudden versus gradual modulations. Sudden modulations tend to put important contrasting features of two maqāmat in immediate juxtaposition. The new mode is usually presented directly after a cadence in the original mode. The contrasting feature of the new maqām is often contained in the new mode’s lower tetrachord. Further, the contrasting feature is often arrived at by an ascending melodic leap. See, for example, the phrases in Example 2 which contain sudden modulations from Bayyātī to Ṣabā, from Ṣabā to Bayyātī, and from Bayyātī to Rāst Nawā.

Gradual modulations, on the other hand, often occur almost inconspicuously in the middle of a phrase (rather than after a cadence). Further, such modulations often begin in the new mode’s upper tetrachord. It is only when the melodic line completes a gradual descent to the lower tetrachord of the new mode that the modulation is fully confirmed (for it is the lower tetrachord which is generally understood to contain a mode’s dominant character). See, for example, the phrases in Example 3 which contain gradual modulations from Bayyātī to Ṣabā, from Ṣabā to Bayyātī, and from Bayyātī to Rāst Nawā.

Example 2: Sudden modulations**Example 3: Gradual modulations**

Another two-fold classification recognizes passing versus full-fledged modulations. The criteria for characterizing a given modulation in these terms include the amount of time one spends in the new mode and whether there are any major cadences in this mode. The presence of even a single strong cadence in the new mode is usually enough for the modulation to be considered full-fledged rather than just passing.⁹

Passing modulations are commonly dismissed with the comment, “This is not a real modulation; it’s just a *ḥarakah*,” that is, roughly “a brief movement.” Recently, the European concept of “passing” modulation has been translated and adopted in academic circles. The idea is rendered in Arabic with the adjective *murūriyyah* (feminine) or *murūrī* (masculine), “passing”; thus *ḥarakah murūriyyah* (“a passing movement”) or *taghyir*

murūrī, *intiqāl murūrī*, and *taḥwīl murūrī* (all meaning “passing modulation”).

Passing modulations play an important role since it is often only in these brief phrases that some of the rarest and least-used modes ever appear in practice. Villoteau quotes from a pre-modern manuscript which attests to this limited use of rare modes in the past (1826:68; quoted above). Example 4 contains a brief passing modulation to maqam Nakriz (C D E $\flat$ F $\sharp$ G A B $\flat$ c) within a Rāst *taqāsīm*:

Example 4: A brief passing modulation



It is also common to label specific examples of modulation as either common/typical or rare/unusual. Tawfīq al-Ṣabbāgh (1950:41ff.) and ‘Abd al-Mun‘im ‘Arafah (1976:21–22), for example, list common modulations for each of the main maqamat. After analyzing sixteen *taqāsīm* performances in maqam Nahāwand by Jihad Racy, Bruno Nettl and Ronald Riddle find

a distinct patterning of modulation usage. The two most commonly used maqamat are Bayyātī Nawā (twelve times) and Rāst (ten). Three others appear with [less] frequency. . . . [Four others] appear only once each. . . . The first modulation is Rāst in eight of the Nahāwand *taqāsīm*, Bayyātī Nawā in six, and Ḥijāz in one. . . . The conclusion regarding the improvisatory decision-making process seems obvious: a performer has in mind a typical sequence [of modulations] (but departs from it occasionally). (1973:19)

The most common modulations are to one or two of the most prominent modes which share the same tonic pitch (especially those which share the same lower tetrachord) and to one or two of the most prominent modes which appear (usually in transposition) based on the first note of the mode’s second tetrachord. From maqam Rāst, for example, the most common modulations are to Sūznāk and Nahāwand (both based on C), to Bayyātī and Ḥijāz, both transposed to G (i.e., Bayyātī Nawā and Ḥijāz Nawā), and to Huzām on E half-flat. From maqam Bayyātī the most common modulations are to Shūrī, Ṣabā, and Ḥijāz (all sharing the same tonic pitch), and to Rāst transposed to G (i.e., Rāst Nawā) (see fig. 6).¹⁰

Figure 6: Common modulations from maqam Rāst:

Rāst:	C	D	E $\flat$	F	G	A	B $\flat$	c				
Sūznāk:	C	D	E $\flat$	F	G	A $\flat$	B	c				
Nahāwand:	C	D	E $\flat$	F	G	A $\flat$	B	c				
Bayyātī Nawá:					G	A $\flat$	B $\flat$	c	d	e $\flat$	f	g
Ḥijāz Nawá:					G	A $\flat$	B	c	d	e $\flat$	f	g
Huzām:			E $\flat$	F	G	A $\flat$	B	c	d	e $\flat$		

Common modulations from maqam Bayyātī:

Bayyātī:	D	E $\flat$	F	G	A	B $\flat$	c	d				
Shūri:	D	E $\flat$	F	G	A $\flat$	B	c	d				
Ṣabá:	D	E $\flat$	F	G $\flat$	A	B $\flat$	c	d $\flat$				
Ḥijāz:	D	E $\flat$	F $\sharp$	G	A	B $\flat$	c	d				
Rāst Nawá:					G	A	B $\flat$	c	d	e	f $\neq$	g

Rare or unusual modulations might involve unusual transpositions (such as a modulation from Rāst to maqam Sikāh transposed to C¹¹), or a direct modulation between two modes that are generally perceived as being distant from one another. One commonly moves to a distant mode by first modulating to intermediary modes which are understood to stand in a closer relationship with the original mode. (This point is discussed further below.) An example of an unusual modulation to a distant mode would be a direct move from Rāst to Ṣabá Nawá. More commonly, one would first modulate from Rāst to Bayyātī Nawá, and then to Ṣabá Nawá (see fig. 7).

Figure 7: Modulating from Rāst to Bayyātī Nawá to Ṣabá Nawá:

Rāst:	C	D	E $\flat$	F	G	A	B $\flat$	c				
Bayyātī Nawá:					G	A $\flat$	B $\flat$	c	d	e $\flat$	f	g
Ṣabá Nawá:					G	A $\flat$	B $\flat$	d $\flat$	d	e $\flat$	f	g $\flat$

Required Modulations?

While the maqamat each have modes to which they most commonly modulate, there are no required modulations in the present day. It was, however, common in the nineteenth and early twentieth centuries for theorists to define some individual modes as a conglomeration of two or more maqamat. These modes, then, contained obligatory shifts from one maqam to another. Mashāqah, for example, describes the mode Kārdānī Ghazalī, “It is the performance of [the mode] Māhūr . . . then you close with [the mode] Bayyātī” ([c. 1840]1899:932). Similarly, Baron Rodolphe D’Erlanger describes a number of compound modes which begin in one maqam and end in another (1949:294–95, 298–301). D’Erlanger also mentions a number

of modes which were understood to contain superimposed tetrachords. He comments that these modes can be considered as containing “simple modulations. These modulations are, however, obligatory; they generally constitute an essential element of the mode and often serve to distinguish it from another” (1949:104). A few present-day theorists continue to mention a small number of these compound or complex *maqamat*. The modes *Ṣabá Zamzamah* and *Ṣabá Busalik*, for example, are occasionally defined as containing modulations between *Ṣabá* and *Kurd*, and between *Ṣabá* and *Busalik*, respectively (see, for example, Surūr 1986:136–37). Similarly, *maqam Zawil* is sometimes mentioned and described as *maqam Rāst* with a touch of *maqam Nakrīz* (Surūr, personal communication, June 1987; see also Muḥammad 1984:37). Most present-day theory books, however, no longer mention such modes.¹²

There is one sense in which modulations might be considered obligatory: when one wants to modulate to what is commonly perceived to be a distant *maqam*. In order to achieve such a modulation, intermediary modulations are generally considered necessary. The specific intermediary steps are not prescribed; however, there might be a consensus on the most common way of achieving a given modulation. Note the example cited above of a modulation from *Rāst* to *Ṣabá Nawá* via *Bayyātī Nawá*.

It is also common for individual musicians to have their own personal approaches to a modulation between two distant modes. Al-Ṣabbāgh, for example, states that “the modulation [*al-intiqāl*] from *Ṣabá* to *Ḥijāz* is not possible except after passing through [an unnamed third mode whose characteristics, he says, are between those of *Ṣabá* and *Ḥijāz*]” (1950:49). As another example, one of my teachers in Cairo stated that when modulating from *maqam Shūrī* to *Rāst Nawá* it is best to insert *Bayyātī* as an intermediary mode (Mansi Amin, personal communication, 25 June 1984) (see fig. 8).

Figure 8: Modulating from *Shūrī* to *Bayyātī* to *Rāst Nawá*:

Shūrī:	D	E ^b	F	G	A ^b	B	c	d
Bayyātī:	D	E ^b	F	G	A	B ^b	c	d
Rāst Nawá:				G	A	B ^b	c	d . . .

By using the concepts of tonic versus non-tonic, sudden versus gradual, full-fledged versus passing, and common versus rare, it is possible to achieve a fairly developed and emically based catalog of the types of modulations that occur in present-day Arab music.

Rules Which Govern Modulation?

For many writers and musicians there are no specific rules which govern the process of modulation. All acknowledge, however, that modulations

must be guided by “good taste” (*dhūq salīm*) and musical sensitivity (see, for example, Ḥāfiẓ 1971:183 and Fahmī 1965:76). It is generally agreed that the ear is the sole judge of whether a particular modulation is correct or not. If a particular modulation is criticized it is said to be “hard on the ear” (*saʿb ʿalā al-widn* in Egyptian dialect).¹³

While many claim that there are no specific rules, others insist that there are clear (but usually unexpressed) principles which govern proper modulation. Al-Shawwā, for example, writes that modulation “has artistic rules [*qawāʿid fanniyyah*] so that by adhering to them and to their conditions order and harmony result rather than disunity and lack of harmony” (1946:96). Mansi Amin, one of my Cairo teachers, stressed this same point (personal communication, 30 May 1987).

The most widely stated rule is that one must return to the original maqam (*al-maqām al-asāsī*) before ending a given piece. This is accepted by writers and musicians who otherwise say that there are no rules. Mashāqah, for example, remarks that if a piece of music “does not return [to the original mode at the end of the piece], it is a defect in the composition” ([c.1840]1899:1081).

The second most widely acknowledged rule is based on the understanding that the modes stand in different levels of proximity to one another. The relationship between any two modes is usually expressed in terms of the adjectives “close,” “closer,” and “distant” (*qarīb* [*qurayyib* in Egyptian dialect], *aqrab*, and *baʿīd*, respectively). In a rare instance where this issue is addressed in the literature, al-Ṣabbāgh expresses the rule of relative proximity:

. . . when improvising [or composing] in whatever mode and one wants to modulate [*yantaqil*] to [another] mode . . . it is necessary that one does not move suddenly from a mode to another distant [*baʿīd*] mode. Rather, one modulates from the mode to the mode which is closest [*aqrab*] to it, then one [can] modulate from this last mode to [a] mode which is close [*qarīb*] to it and in this manner one modulates step by step until one reaches a mode which is completely distant from the original mode in which one was improvising [or composing] and which, if one had modulated to it suddenly, would have created an explosion in the ears of the listeners. . . . After this, whenever one wants to return to the original mode, one [must] do so by the same . . . process, that is, return step by step from one mode to the mode which is the closest to it until one reaches the original mode. (1950:54)

Mansi Amin expressed the same idea, stating, “It is necessary that one knows the *qarāyib*” (pl. of *qurayyib*), that is, the maqamat which stand in close relationship with the maqam in which one is improvising or composing (personal communication, 25 June 1984).

Relative proximity is based on two points; first, the degree of structural and scalar affinity between any two modes, and second, on the degree of

compatibility which is understood to exist between the modes' tonic pitches. The greatest level of proximity is found between modes which share the same lower tetrachord (modes of the same *faṣīlah*). Rāst and Sūznāk are an example of this relationship. Modes which share the same upper tetrachord but differ in their lower tetrachord are considered close, but not as close as those of the same *faṣīlah*. For example, Sūznāk, Huzām, and Nahāwand all share a Ḥijāz tetrachord on G (see fig. 9).

Figure 9: Modes which share the same lower tetrachord:

Rāst:	C	D	E♭	F	G	A	B♭	c
Sūznāk:	C	D	E♭	F	G	A♭	B	c

Modes which share the same upper tetrachord but differ in their lower tetrachord:

Sūznāk:	C	D	E♭	F	G	A♭	B	c
Huzām:			E♭	F	G	A♭	B	c d e♭
Nahāwand:	C	D	E♭	F	G	A♭	B	c

Sūznāk and Huzām are generally understood to stand in closer relationship to one another than Nahāwand and Huzām since the former pair share the note E half-flat (the tonic in Huzām, a secondary *ghammāz* in maqam Sūznāk).

More important than structural or scalar similarities is the degree of compatibility understood to exist between two modes' tonic pitches. The overriding nature of tonic compatibility can be seen in the following examples of modes which share the same notes and the same upper tetrachord but which, nevertheless, are considered distant from one another because of "tonic incompatibility." Such is the case with the modes Rāst and Bayyātī, and Sūznāk and Shūrī, where a general "prohibition" against modulations between C-based and D-based maqamat renders these two pairs of modes entirely distant from each other. The same is true for the modes Shūrī and Huzām, where a "prohibition" against modulations between D-based and E♭-based maqamat renders these two modes completely distant from one another in terms of modulation (see fig. 10).

Figure 10: Examples of "distant" modes:

Rāst:	C	D	E♭	F	G	A	B♭/B♭	c
Bayyātī:		D	E♭	F	G	A	B♭/B♭	c d
shared upper tetrachords ¹⁴								
Sūznāk:	C	D	E♭	F	G	A♭	B	c
Shūrī:		D	E♭	F	G	A♭	B	c d
shared tetrachord								

Shūrī:	D	E♭	F	G	A♭	B	c	d	
Huzām:		E♭	F	G	A♭	B	c	d	e♭
				└──────────┘					
				shared tetrachord					

The concept of tonic compatibility has as its basis the commonly held classification of the maqamat according to tonic pitch.¹⁵ In this system of classification the main maqamat fall into one of five different groups (see fig. 11).¹⁶

Figure 11: The main maqamat classified by tonic pitch:

the C maqāmāt:	Rāst	the D maqāmāt:	Bayyātī
	Sūznāk		Shūrī
	Nahāwand		Şabā
	Hijāz Kār		Kurd
	Hijāz Kār Kurd		Hijāz
	Nawā Athar		Shahnāz
	Nakrīz		
the E♭ maqāmāt:	Sīkāh		
	Huzām		
the F maqām:	Jahārkāh		
the B♭ maqām:	‘Ajām ‘Ushayrān		

Modulatory practice has a grammar which is based, in part, on each of the above maqam groups acting as discrete entities. Membership in one or the other of the above “families” tends to determine the possible modulations to and from a given maqam, the overriding factor being tonic compatibility. Using these “families” of maqamat as a point of departure, I have formulated four rules which explain the vast majority of the modulations that occur in modern-day practice. These are the rules for (1) tonic modulations, (2) modulations between the two primary families, the C and D maqamat, (3) modulations to and from the E half-flat and B flat maqamat, and (4) modulations to the F maqam(at). These rules are based on the teachings I received from Jihad Racy in the United States and from numerous teachers in Cairo. They are also based on extensive analysis of pieces in the modern repertoire.

The repertoire I am referring to includes both the “musically learned” and the “popular” domains. The former is represented by the *samā’ī* and *bashraf* genres, the latter by the modern song tradition (including the songs of Umm Kulthūm, Muḥammad ‘Abd al-Wahhāb, Farīd al-Aṭrash, ‘Abd al-Ḥalīm Ḥafīz, Wardah, Fayrouz or Fairūz, etc.).¹⁷ Modulations in the “musically learned” and “popular” domains of Arab music have, it seems, been governed by the same body of rules since the second half of the nineteenth century and possibly earlier. (Few pieces from before the second half of the

nineteenth century survive.) The rules presented below are followed virtually without exception in this body of music. After presenting the four rules, I contrast the modulatory grammar that exists in this music with the grammar that prevails in folk music domains.

Rule #1: Tonic Modulations

The first rule can be stated: modulation occurs freely among maqamat which share the same tonic.

Thus, in a D maqam, one can modulate to any other D maqam; in a C maqam, one can modulate to any other C maqam; etc.¹⁸ Tonic modulations might make up as much as 50% of the modulations found in Arab music. "*Samā'ī* Bayyātī al-Thaqīl," transcribed above, provides examples of modulations from Bayyātī to Ḥijāz (mm. 12–13) and Šabā (mm. 16–18), all three being D maqamat.

Rule #2: Modulation Between the Two Primary Families

A review of the maqamat listed above reveals that the C and D families make up the main portion of the maqam system. Modulation between these two families is restricted by the following rule: one cannot modulate from a C-based maqam to a D-based maqam or vice versa.

It would seem then that the two families are mutually exclusive. However, this is where transposition comes into play: to modulate from a C maqam to any D maqam, transpose the D maqam to G, and modulate to the transposed maqam. Similarly, to modulate from a D maqam to any C maqam, transpose the C maqam, also to G, and modulate to the transposed maqam. An example of this latter situation occurs in "*Samā'ī* Bayyātī al-Thaqīl" where maqam Rāst is transposed to G (mm. 8–9).¹⁹

Rule #3: Modulation to the E half-flat and BB Maqamat

The E half-flat maqamat are treated as if they were members of the C family. Thus, modulations between the C and the E half-flat maqamat occur freely as if they were simple tonic modulations. See, for example, "*Al-Amal*," composed by Zakarīā Aḥmad and sung by Umm Kulthūm (Sono Cairo cassette 76006), the first verse of which contains a direct modulation from Rāst to Huzām.

Modulations between the D and E half-flat maqamat require the same transpositions as modulations between the C and D maqamat: either the D maqamat are transposed to G (as stated in rule #2 above) or the E half-flat maqamat are transposed to B half-flat (or BB half-flat). The latter point is explained as follows: Sikāh and Huzām normally start a neutral third above C; after transposition they would start a neutral 3rd above G, i.e., B half-flat (see fig. 12).

Figure 12: Rāst with its Sīkāh/Huzām:

Rāst on C: C D E $\flat$ F G ...
 its Sīkāh/Huzām: E $\flat$ F G ...

Bayyātī with its Rāst on G and its Sīkāh/Huzām on B half-flat:

Bayyātī on D: D E $\flat$ F G ...
 its Rāst on G: G A B $\flat$ c d ...
 its Sīkāh/Huzām: B $\flat$ c d ...

See, for example, the introduction to the Umm Kulthūm song, “Wa Marrat al-Ayyām” (Sono Cairo cassettes 76040 or 81299), which contains a modulation from Bayyātī on D to Huzām transposed to BB half-flat. (This mode is often called Rāḥat al-Arwāḥ.) Another Umm Kulthūm song, “Inta ‘Umrī” (Sono Cairo cassette 01), contains a similar modulation transposed down a fourth, from Kurd on AA to Huzām on F half-sharp.

The BB $\flat$ maqamat are treated as if they were members of the D family. Thus, modulations between the D maqamat and ‘Ajam ‘Ushayrān, the main BB $\flat$ maqam, occur freely as if they were simple tonic modulations. Modulations from the C maqamat, however, require that ‘Ajam ‘Ushayrān be transposed to E $\flat$ (or possibly to C).

‘Ajam ‘Ushayrān’s transposition to E $\flat$ is explained as follows: this mode normally starts a major third below D; after transposition it would start a major third below G, on E $\flat$. This transposition occurs only rarely. If ‘Ajam ‘Ushayrān were transposed to C, then it could be introduced as a tonic modulation from other C-based modes such as Rāst.

Figure 13: Bayyātī with its ‘Ajam:

Bayyātī: D E $\flat$ F G A B $\flat$ c d
 its ‘Ajam: BB $\flat$ C D E $\flat$ F G A B $\flat$

Rāst with its Bayyātī and its ‘Ajam

Rāst on C: C D E $\flat$ F G A B $\flat$ c
 its Bayyātī: G A $\flat$ B $\flat$ c d e $\flat$ f g
 its ‘Ajam: E $\flat$ F G A $\flat$ B $\flat$ c d e $\flat$

The foregoing three rules probably account for more than 90% of the modulations that occur in modern-day Arab music.

Rule #4: Modulation to the F Family of Maqamat

A review of the above patterns of modulation and transposition reveals that the maqam system revolves primarily around two series of axes. The first is C E $\flat$ G (and E $\flat$). This is because the C and E half-flat modes act as one family by freely accepting modulations from each other and by both requiring that

the D modes be transposed to G and the BB $\flat$ modes be transposed to E $\flat$. The second is D G B $\flat$ /BB $\flat$ (and BB $\flat$ /B $\flat$). This is because the D and BB $\flat$ modes also act as one family by freely accepting modulations from each other and by requiring that the C modes be transposed to G and the E half-flat modes be transposed to B half-flat (or BB half-flat).

These two systems of axes have the note G in common. G's role is explained, in part, by two factors. First, as noted above, G is the base note of the upper tetrachord (the *ghammāz*) for most C, D, and E half-flat modes. This is because most C maqamat are perceived as being made up of disjunct tetrachords (C - F and G - c), most D maqamat are conceptualized as containing conjunct tetrachords (D - G and G - c), and the E half-flat maqamat are seen as being formed by an initial trichord followed by a conjunct tetrachord (E $\flat$ - G and G - c). Thus their second tetrachords all start on G. When this situation is combined with a second factor, namely, that non-tonic modulations most commonly occur on the first note of a mode's second tetrachord, then G is rendered an extremely important note in the realm of modulation.²⁰

How then do the F maqamat fit into the maqam system, for they seem to be excluded from the two sets of axes mentioned above? F modes interact with the other modes in one of three ways. First, they can appear in modulations from the BB $\flat$ modes since BB $\flat$ modes are commonly analyzed as having a second tetrachord that starts on F (BB $\flat$ - E $\flat$ and F - B $\flat$). There are, however, few pieces based in the BB $\flat$ modes (of which only 'Ajām 'Ushayrān is commonly mentioned). A second way that F modes appear is when F is treated as a secondary *ghammāz* within C- or D-based maqamat. Thus, one might modulate from Bayyātī or Rāst to Jahārkhāh, the main F maqam. This possibility also occurs only rarely. A third link between the F modes and the rest of the maqamat is provided by two C maqamat which function, at times, as if built from conjunct instead of disjunct tetrachords (C - F and F - B $\flat$). During such moments, these modes (Hijāz Kār Kurd and Hijāz Kār) accept modulations to the F maqamat and also to some of the C maqamat transposed to F.²¹ See, for example, the Umm Kulthūm song, "Awadt 'Aynī" (Sono Cairo cassette #75010), which contains modulations from Hijāz Kār Kurd to Rāst on F and Nahāwand on F.

The preceding discussion codifies modulatory practice in both the "musically learned" and "popular" domains. There is, however, a conflicting practice within Egyptian folk music, specifically in the *madīḥ al-nabī* (or *madḥ*) and *mizmār/taḥl baladī* traditions which are performed at saints-day festivals (*mawālīd*; singular *mawlid*, pronounced *mūlid*) and at numerous other occasions where folk entertainment is required (weddings, birthdays, etc.). *Madḥ* is also performed at folk *dhikr*s (pronounced *zīkr*). In these performances an uninterrupted suite of *mawāwīl* (singular *mawwāl*) and other vocal genres commonly start with the C maqamat (using any number

of the C maqamat), then move up the scale to the D maqamat (again using any number of these maqamat), and then close with the E half-flat maqamat. See, for example, a performance by Shaykh Farid Hagāg (*Sabrah fī Madh al-Rusūl*, Sawt al-Hilāl cassette) which contains modulations from C to D and from D to E half-flat modes.²²

This stepwise progression up the scale is not allowed in the more “mainstream” traditions, that is, in the “musically learned” and “popular” domains. Having found a tradition which does not follow the restrictions discussed above, we can now view these restrictions in a new light. They are not inherent to the maqam system. Rather, it might be argued that they are an attempt to create a more refined, a more highly evolved modal system, one that stands in contrast to traditions found in (at least some of) the region’s folk musics.

Performance Strategies

The foregoing rules address the issue of which modulations are possible and which are not. In addition, we can also study the processes by which specific modulations are achieved in practice. In their compositions and improvisations, musicians tend to follow specific strategies when performing given modulations. In contrast to the above rules, however, which are seldom stated explicitly, these strategies are often discussed with some detail. For example, it is somewhat common for musicians, when demonstrating modulations, to pinpoint one or two specific notes from which they begin a given modulation. One teacher referred to these notes informally as “the door” (*al-bāb*) to a modulation (Mansi Amin, personal communication, on numerous occasions in 1982–83, 1984, and 1987). Traditional Arab music theory, however, does not recognize either this concept or the phenomenon as a whole.

For modulations among many of the D-based modes (Bayyātī, Šabā, Kurd, and Hījāz) and ‘Ajām ‘Ushayrān (based on B♭) B♭ often serves as this pivot note. When composing or improvising, one can modulate among these modes by first focusing on B♭ using the note A as a leading tone to the B♭. Then, by descending with B♭ A G F E♭ D, one has shifted to maqam Bayyātī; by descending B♭ A G♭ F E♭ D, one has moved to Šabā; B♭ A G F E♭ D C B♭ and one has entered maqam ‘Ajām ‘Ushayrān, and so on (see fig. 14).

Figure 14: The note B♭ as a pivot note in a number of maqamat:

Bayyātī:	D	E♭	F	G	A	B♭	c	d
Šabā:	D	E♭	F	G♭	A	B♭	c	♭
Kurd:	D	E♭	F	G	A	B♭	c	d
Hījāz:	D	E♭	F♯	G	A	B♭	c	d
‘Ajām ‘Ushayrān:	B♭	C	D	E♭	F	G	A	B♭

In modulations between two modes which have dissimilar tetrachords based on the same note, the pivot note is often one of the two notes which define the boundaries of the dissimilar tetrachords. For example, modulations from Rāst to Bayyātī Nawá often use either G or c as a pivot note. (Maqam Rāst has either a Rāst or Nahāwand tetrachord on G; Bayyātī Nawá has a Bayyātī tetrachord on G.)

Figure 15: The modes Rāst and Bayyātī Nawá both have tetrachords based on G:

Rāst:	C	D	E♭	F	G	A	B♭/B♮	c				
Bayyātī Nawá:					G	A♭	B♭	c	d	e♭	f	g

A number of musicians indicated that one can first hold the note c within maqam Rāst and then modulate to Bayyātī Nawá (see Example 5).

Example 5: A modulation from Rāst to Bayyātī Nawá using the note 'c' as a pivot note



By indicating such pivot notes, musicians are not limiting or restricting the ways a particular modulation can be achieved. Rather, they are pinpointing specific strategies they use for achieving modulations which are “acceptable to the ear.” (For other examples of pivot notes, see the examples of gradual modulations given above: the note d♭ begins the gradual modulation from Bayyātī to Šabá, d begins the shift from Šabá to Bayyātī, and the note e begins the gradual modulation from Bayyātī to Rāst Nawá.)²³

After fieldwork in 1798–1801, Villoteau reported finding rules for modulation based on a concept akin to pivotal tetrachords:

In Arab music, as in ours, there are rules for passing from one mode to another, of which the principle is to always announce by some preparatory notes which prepare the ear for the change of the [mode] which is about to take place. This preparation consists chiefly of the interlinking of the notes of the [tetrachord] where . . . one wants to make the transition, with those of the analogous and corresponding [tetrachord] of the mode in which one wants to enter. In this manner one can pass successively among all the modes, and return to that from which one has come without offending the ear. (1826:126–27)

In conclusion, we have examined a phenomenon that lies largely outside the domain of written Arab music theory. Modulation is, neverthe-

less, observable and thus able to be documented in a number of different realms: in musicians' improvisations, in precomposed repertoire, in prevailing oral theories, in music lessons, and in occasional remarks found in historical treatises. When this wealth of information is compared to the virtual absence of comments on the subject in modern Arabic theoretical literature, we are forcefully reminded once again of the gulf that can separate a music tradition from its parallel written theoretical tradition. What is of great importance to the musician might, historically speaking, be of little importance to the music theorist and vice versa. The present study thus emphasizes the importance of new scholarship which reexamines the existing boundaries of theoretical traditions. Such studies can be especially efficacious when musicians' oral concepts and performance rules and strategies are examined from an etic perspective.

Notes

1. Research in Cairo was supported by a grant from the American Research Center in Egypt (ARCE), funded by the Smithsonian and ICA (Fulbright). I would like to thank A. Jihad Racy, Nazir A. Jairazbhoy, Dwight Reynolds, and my teachers and colleagues in Cairo for their knowledge, comments, and support.

In this article, I use "mode" and *maqām* interchangeably. *Maqām* can refer to either a simple or a very complex set of phenomena. In its simplest form, it may refer to a specific scale. Defined in greater depth, *maqām* may refer to a specific scale with a tonic, alternative directional notes, accidentals, a specific ambitus, a specific tetrachordal structure with alternative tetrachordal structures, one or more prominent notes besides the tonic, starting notes, specific paths for performance, melodic motives, specifics of intonation, and extra-musical associations. These and other issues are discussed in Marcus 1989:438–754. The plural of *maqām* is *maqāmat*.

2. Except for dictionary definitions (Wehr 1976), all translations from Arabic and French are my own.

Harold Powers (1980:426, 427) states that the process of changing from one scale type to another is called *tarkīb* (see also Pacholczyk 1980:523). This usage might be based on D'Erlanger 1949:101–104. However, a review of nineteenth- and twentieth-century Arabic literature and interviews with present-day theorists and musicians confirm that the word *tarkīb* is not a recognized term in the "modern period." *Tarkīb* was a musical term in the medieval and late-medieval periods (see, for example, al-Faruqi 1981:355).

The "modern period" is defined as beginning with the conceptualization of the 24-note quarter-tone scale. This is generally thought to have occurred in the early to mid-1700s. See Marcus 1989 for a detailed discussion of the concept of the "modern period."

3. The question is often raised whether modulation is more common today than in the past. Unfortunately, the sources used for the present study do not offer any evidence which supports an answer one way or the other. The system of levers ('*urab*') which were applied to the *qanūn* in the first decades of the twentieth century (see Marcus 1989:240–42), thereby greatly facilitating the execution of modulations on this instrument, remains the major evidence in support of the idea that modulation is more prevalent in the twentieth century than it has been in the past. However, Alfred Berner documents modulatory techniques on the *qanun* prior to the application of the '*urab*' (1937:27–30).

4. When I arrived in Cairo I expected to focus on how musicians conceptualize melodic movement within the individual modes. However, apart from the general ascending progression (see Marcus 1989:698–703), my teachers offered little instruction about movement within a given mode. When I felt I had exhausted the possibilities for learning about conceptualization

within a maqam, I began to ask about modulation. The reaction to my new focus was remarkable: my teachers seemed to let out huge sighs of relief. One of my *ūd* teachers, George Michel, summed up the general feeling: "Ah, now *this* I can help you with. Yes, *this* is my business" (13 October 1982). Indeed, for the rest of my time in Cairo, maqam lessons with Michel and a number of other teachers focused primarily on modulation.

5. In classes at UCLA, Jihad Racy refers to these as "tonic" and "relative" modulations, respectively. See Nettl and Riddle 1973:18 where Racy's two-fold classification is mentioned.

6. See Marcus 1989:377–96 for an extensive introduction to the *faṣīlah* concept. Sāmī al-Shawwā used the word *faṣīlah* to refer to a group of modes which shares the same tonic, irrespective of the structure of the modes' lower tetrachords; this usage predates the present-day understanding of the term. Thus, when al-Shawwā writes that modulation is permitted among maqamat of the same *faṣīlah*, he gives as examples modulations from maqam Rāst to Sūzdulār, Sūznāk, Nahāwand, and Nawā Athar, all C-based modes (1946:98). According to the present-day understanding, Rāst, Nahāwand, and Nawā Athar belong to different *faṣā'il* because of their unique lower tetrachords.

Amīn Fahmī refers to modes which share the same tonic as *aqārib* ("relatives") (1965:76). He adds that one can modulate freely among *aqārib*. He acknowledges that he has borrowed this term from the Western concept of "relative" keys.

7. Surūr defines *ghammāz* as "a note from which you change either the tetrachord [*jins*] or the maqam" (personal communication 18 June 1987). See similar definitions in Surūr 1986:117 and Muḥammad 1984:21.

8. A few present-day theorists, seeking to develop a "scientific" approach to discussions of modulation (scientific but not empirically based), have allowed the possibility of modulations to every degree of a given mode's scale, including the second and seventh. See, for example, Shurāh (1984) who admits modulations from the C-based maqam Rāst to a number of D-based maqamat. Surūr also holds that there are no restrictions regarding modulation to a given mode's various scalar degrees (personal communication 19 May 1987). He does, however, recognize that modes based on D and B half-flat stand in a "distant" relationship with maqam Rāst, whereas modes based on G, E half-flat, and A stand in a "close" relationship. (The concepts of "close" and "distant" modes are discussed below.) Gerson-Kiwi writes about maqam Rāst that "every intervallic degree belonging to Rāst [can develop] into a nearly independent maqam centre propelling out of its root additional side-maqamat" (1970:72).

9. It is not necessary that the complete octave scale of the new mode be used for the modulation to be considered full-fledged. A modulation from Rāst to Bayyātī Nawā (G A^b B^b c d e^b f g), for example, might give only the notes F G A^b B^b and c of the latter mode (see Powers 1980:425). Karl Signell discusses passing modulations with respect to Turkish music practice (1977:77ff.).

10. Modulations to Bayyātī and Rāst (whether in root position or transposed to G) are common for most maqamat. Al-Ṣabbāgh, for example, comments that maqam Bayyātī is a prominent maqam which enters into all the other modes (1950:47).

11. It is very unusual for modes based on half-flat positions to be transposed to natural-note positions. This is because the resultant scales have a large number of half-flat and half-sharp notes and are considered difficult to perform with correct intonation. Maqam Sīkāh, for example, is based on E half-flat (E^b F G A B^b c d e^b). When transposed to C it would have the following notes: C D^b E^b F[♯] G A^b B^b c.

12. Tawfiq al-Ṣabbāgh, a Syrian violinist and music theorist, speaks against the continued recognition of such compound or complex modes (1950:35; quoted in Marcus 1989:685).

13. Villoteau found this to be the case in Egypt in 1798–1801. He reports that modulations must be performed "without offending the ear" (1826:127; see fuller passage quoted below).

14. The modes Rāst and Bayyātī use both B flat and B half-flat, and thus have alternative upper tetrachords of G A B^b c and G A B^b c. These are called Nahāwand and Rāst tetrachords, respectively.

15. The concept of "tonic compatibility" is my own and thus there is no equivalent term in modern Arab music theory. Note, however, the use of *qarīb* and *ba'īd* described above. My

presentation of the concept is based on teachings I received in Cairo from George Michel, Mansi Amin, and numerous other scholars and musicians.

16. See Marcus 1989:368–77 for further discussion of this classification system. Maqamat based on GG (such as Farahfāzā, Shatt ‘Arabān, and Yakāh) and AA (Bayyātī ‘Ushayrān, etc.) are not included here as they are seldom played in the present day. When they do occur, they are treated as transpositions of similar C- and D-based modes. BB half-flat maqamat such as ‘Irāq and Rāḥat al-Arwāḥ are treated here as transpositions of E half-flat maqamat.

17. Racy refers to the latter as the “central domain” (1981:12): “This music is . . . known even among non-Arabs as the dominant style of ‘Arab music’ ” (Racy 1982:391).

18. See note 6 above.

Racy offers two exceptions to this rule of unrestrained tonic modulations (personal communication, 3/1986). The cases of Ḥijāz Kār and Ḥijāz Kār Kurd, two C-based modes, are discussed below in rule #4.

19. Mansi Amin, one of my teachers who stated that modulations are not possible between C- and D-based modes, allowed two exceptions. He allowed modulations from the C-based Nawā Athar and Nakrīz to D-based modes and vice versa. This, he explained, is because of the strong presence of a Ḥijāz tetrachord on D in these two C-based modes (personal communication, 20 June 1987). Others denied these exceptions, in part, by refusing to recognize the existence of this Ḥijāz tetrachord; they acknowledge only an initial pentachord in these modes.

Nawā Athar:	C	D	E ^b	F [#]	G	A ^b	B	c
Nakrīz:	C	D	E ^b	F [#]	G	A	B ^b	c

It is interesting to note that in pieces of music using this same set of notes, Greek music practice allows frequent shifts from a C tonic to a D tonic. This shows that there is nothing inherent in the modal system itself which would, by necessity, disallow such modulations in Arab music. Rather, this restriction is an aesthetic choice made by Arab music practitioners.

20. Even maqām Ṣabā, whose tetrachordal structure differs from the norm for D-based modes (it has G^b instead of G and is commonly analyzed having a diminished tetrachord on D followed by an overlapping tetrachord on F: D E^b F G^b + F G^b A B^b) functions as the other D maqamat in terms of modulation. That is, it accepts the C maqamat by transposing them to G (for example, Ṣabā to Rāst on G).

21. During such moments, Ḥijāz Kār would be analyzed as containing a tetrachord on C followed by a conjunct pentachord on F. Ḥijāz Kār Kurd would be analyzed as having conjunct tetrachords on C and F. At other times these two modes function as if they contain disjunct tetrachords on C and G.

Ḥijāz Kār:	C	D ^b	E	F	G	A ^b	B	c
Ḥijāz Kār Kurd:	C	D ^b	E ^b	F	G	A ^b	B ^b	c

22. I thank George Sawa for pointing out the stepwise modulatory practice within the *mizmār/ṭabl baladī* tradition. The *mizmār* is a double reed, oboe-type instrument which is played with circular breathing. The *ṭabl baladī* is a large double-headed drum.

23. Gerson-Kiwi notes the function of pivot notes when she writes that related modes have “a few identical degrees of their modal scales which serve as a platform for mutual excursions. The [*taqāsīm*] player . . . has to seize such a narrow bridge of common notes as a starting point for the transfer to a melodically related *maqām*” (1970:71).

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